

ABSTRACT

This paper introduces DataFITS (Data Fusion on Intelligent Transportation System), an open-source framework that collects and fuses traffic-related data from various sources, creating a comprehensive dataset. We hypothesize that a heterogeneous data fusion framework can enhance information coverage and quality for traffic models, increasing the efficiency and reliability of Intelligent Transportation System (ITS) applications. Our hypothesis was verified through two applications that utilized traffic estimation and incident classification models. DataFITS collected four data types from seven sources over nine months and fused them in a spatiotemporal domain. Traffic estimation models used descriptive statistics and polynomial regression, while incident classification employed the k-nearest neighbors (KNN) algorithm with Dynamic Time Warping (DTW) and Wasserstein metric as distance measures. Results indicate that DataFITS significantly increased road coverage by 137% and improved information quality for up to 40% of all roads through data fusion. Traffic estimation achieved an R^2 score of 0.91 using a polynomial regression model, while incident classification achieved 90% accuracy on binary tasks (incident or non-incident) and around 80% on classifying three different types of incidents (accident, congestion, and non-incident).